

Two Fences

A Reader's Companion to *Radial and Angular Risk in Stochastic DC Optimal Power Flow*

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This is the same paper you already have, rebuilt out of objects you can hold. Every proposition in the formal paper appears here in the same order and with the same number, so you can read the two side by side. Nothing is watered down—the rigour is identical. Only the packaging changed.

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1 First: how to read a paper like this

The formal paper is written in a house style that mathematicians use because it compresses well, not because it is trying to be difficult. Here is the whole vocabulary in one table.

Word	What it actually means
Definition	“Here is a name I am about to reuse fifty times. Pin it down now so I never have to re-explain it.” Carries no claim—you cannot disagree with a definition, only find it useless.
Proposition	A claim that is true, and here is why. The workhorse.
Theorem	A proposition the author thinks is the important one. Same logical status; more self-importance.
Lemma	A small true thing proved only because a bigger proof needs it. A stepping stone.
Corollary	“And therefore, almost for free, this too.” It falls out of the thing just proved with barely any extra work.
Remark	A comment. Often the most useful part, because it is where the author admits what worried them.
Proof	The argument. Ends with $\square$, which just means <i>done</i> .
iff	“if and only if”—the claim runs in both directions. Not a typo.

How to actually read one. Not front to back. Do this instead:

1. Read the abstract. Then read the conclusion. Now you know where it lands.
2. Read every **Definition** and the *statement* of every **Proposition**, skipping all proofs. That is the whole skeleton, and it is usually four pages, not eighteen.
3. Only now go back for the proofs, and only for the results you doubt.

A proof is not there to convince you the author is clever. It is there so that if you *don't* believe the claim, you have somewhere to push.

2 The one picture that carries the whole paper

If you keep one thing from this document, keep this.

You are standing in an open field at a spot marked **zero**—no neighbourhood on the island is asking for any power at all.

Now demand starts to grow, and you **walk**.

- **How far you walk** = how much total power the island is asking for. A hundred metres out might be 600 MW; four hundred metres out might be 1100 MW.
- **Which way you face** = the *mix*. Walking north-east might mean “Honolulu wants a lot and Waianae wants a little.” Walking due south might mean the opposite.

Two fences are drawn on this field, and they are **not the same shape**.

The inner fence is round. It sits about 585 MW out in every direction. Cross it and some transmission line somewhere hits its limit. This is the *congestion* fence.

The outer fence is lumpy. In some headings it sits 1496 MW out; in others it comes

in as close as 409 MW. Cross it and the grid physically cannot deliver—you have to switch customers off. This is the *adequacy fence*.

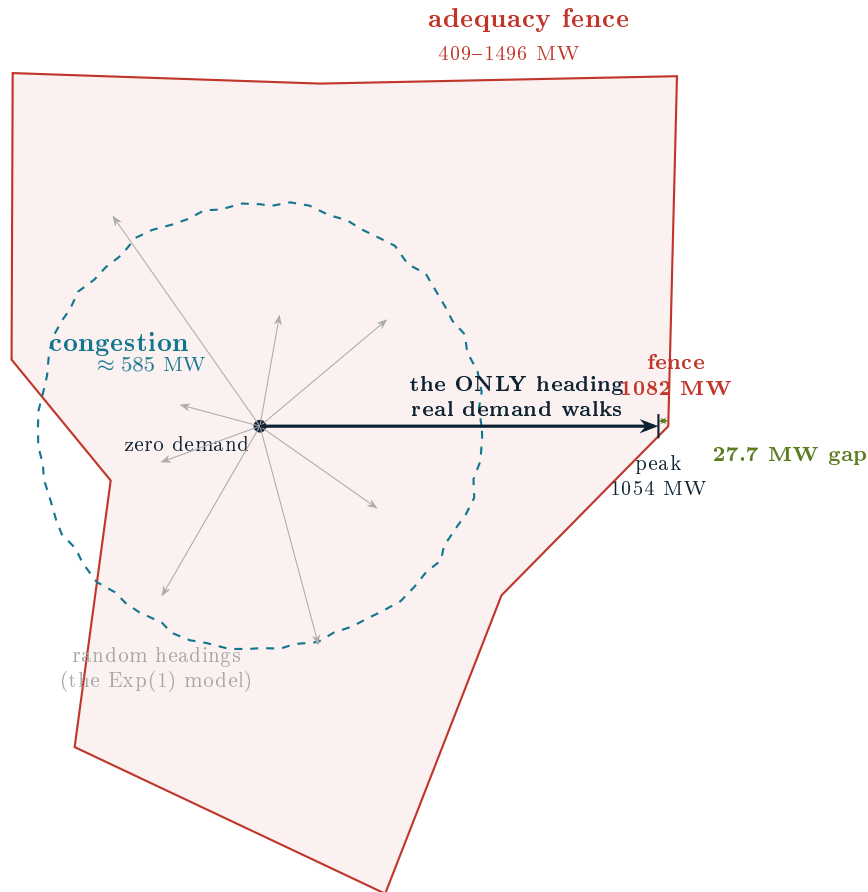


Figure 1: The two fences. Distance from the centre is *how much* power the island wants; the direction you face is *which neighbourhoods* want it. Congestion is a near-circle—it barely cares about direction. Adequacy is a lumpy polygon—it cares enormously. Real O’ahu demand only ever walks one heading (bold), stopping 27.7 MW short of the fence. Note the heading near the bottom-left where the lumpy fence dips *inside* the round one: in a few unlucky mixes the grid runs out of delivery before the network is even fully stressed. This is a two-dimensional shadow of an eight-dimensional picture—there is one axis per load bus—but every feature drawn is real.

Everything else in the paper is a consequence of Figure 1:

What the picture shows	Which result says it
The servable region has no holes: walk out along a heading and you are fine, fine, fine, then never fine again.	Proposition 4.1
Whether you shed depends on <i>which way you face</i> , because the outer fence is lumpy.	Corollary 4.2
Whether you congest barely depends on facing at all, because the inner fence is round.	Proposition 4.5
Real demand always walks one heading, and the fence on that heading is 1082 MW while demand tops out at 1054 MW.	Theorem 4.6

3 The machine (paper §2)

3.1 The rig

A plumbing rig bolted to a wall.

- **Pumps** = generators. Each has a maximum it can push and a price per litre.
- **Pipes** = transmission lines. Each has a maximum it can carry.
- **Taps** = the neighbourhoods. Each is opened to some position; that is the demand.
- **Pressure** = voltage phase angle. Water runs from high pressure to low, and how much runs down a pipe is set by the pressure *drop* across it. You never choose the flows directly—you choose the pressures, and the flows follow.

Your job: open the cheapest pumps needed to satisfy every tap, without overloading any pipe.

That is the linear program labelled (1) in the paper. Piece by piece:

Symbol	Object	Its job
P_g	how hard pump g is running	what you choose
$p_g^{\max}$	the pump's rating plate	you cannot exceed it
c_g	the price tag on that pump	MW ⁻¹ h
θ_i	pressure at junction i	what you really choose
F_ℓ	flow down pipe ℓ	follows from the pressures
$\bar{F}_\ell$	the pipe's diameter rating	flow can't exceed it
D_i	how far tap i is opened	the random part
S_i	water that spills out the drain at tap i	unserved demand

And the objective, equation (1a):

$$\min \underbrace{\sum_g c_g P_g}_{\text{the fuel bill}} + \underbrace{\text{VOLL} \sum_i S_i}_{\text{the fine for spilling}}$$

3.2 Proposition 2.4 — the rig can never break

The claim: *no matter how the taps are set, there is always a valid setting of the rig.*

Turn every pump **off**. Let every tap spill entirely down its drain.

Nothing is being delivered, and the fine is enormous—but look at the rig: no pump exceeds its plate (they're off), no pipe exceeds its rating (nothing is flowing), and the books balance at every junction (zero in, zero out). It is a *terrible* answer. It is a *valid* answer.

That is the entire proof. In this kind of mathematics, to prove “a valid setting exists” you do not reason about all possible settings—you **point at one**. Finding an ugly one is

completely legitimate.

Why bother proving something so obvious? Because without it the whole study is built on sand.

If the rig could break, then a run where the solver returns **INFEASIBLE** is *ambiguous*: did the island genuinely run out of power, or did the solver just trip over a rounding error? You'd be counting two different things in the same bucket. **Proposition 2.4 guarantees the solver never has an excuse**, so every failure that appears is a real one, and it arrives as a number of megawatts rather than a shrug.

3.3 Proposition 2.5 — how big must the fine be?

Making the rig unbreakable creates a new problem: now it can spill *by choice*. If the fine is low, the optimiser will cheerfully cut off a neighbourhood to avoid firing an expensive pump—and you would record that as a blackout when it was really a purchasing decision.

So: how high must the fine be?

The obvious guess is “**higher than the most expensive pump.**” That guess is wrong, and the reason is worth holding onto.

Picture a tap at the far end of a pipe that is already completely full. That neighbourhood cannot get one more litre *at any price*, because the pipe is the constraint, not the pumps. If you could somehow teleport one litre in, what would it be worth there? Far more than the priciest pump charges—because the scarcity is the pipe, and the pipe is not for sale.

In our own data this is not hypothetical: the Part 4 study recorded a price of 381 MW^{−1} h at a bottled-up bus, against a most expensive generator of 237.97 MW^{−1} h.

The quantity you must beat is λ_i^o —the **value of one more megawatt at bus i** , which economists call the locational marginal price and mathematicians call a *dual*. The condition is equation (9):

$$\text{VOLL} > \max_i \lambda_i^o.$$

We set $\text{VOLL} = 10,000 \text{ MW}^{-1} \text{ h}$ —about 42× the costliest generator—and then **checked it held** on 6000 replications. Every time the model shed load, an independent test confirmed that serving that demand was genuinely impossible. No exceptions.

The one piece of machinery in the proof. The proof leans on a shape, so let me give you the shape.

Let $V(u)$ be *the cheapest possible fuel bill to serve demand u* . Plot it. It is **bowl-shaped**: as you demand more, cost rises, and it rises ever more steeply, because you are climbing the merit order into worse and worse pumps.

Now take a **straight ruler** and slide it up underneath the bowl until it just touches at one point. The ruler lies below the curve everywhere and touches it once. That ruler is a **subgradient**. Its slope is λ° —the price of one more megawatt right there.

The proof then says, in one line: shedding a megawatt saves you at most what the ruler’s slope promises, and costs you VOLL. If VOLL beats every ruler slope, shedding is always a losing trade. Hence it only ever happens when serving the demand was impossible to begin with. $\square$

Explain to a friend, using the full-pipe picture and no symbols, why “set the penalty higher than the most expensive generator” is not a safe rule.

4 Rolling the dice (paper §3)

Three ways to decide how far the taps get opened.

	Name in the paper	What it does
A	independent Exp(1)	Every tap rolls its own die, nobody coordinates. The assignment’s starting point.
B	common-shock mixture	One island-wide shock plus private noise at each tap. Closer to physics: one weather system.
C	empirical resampling	No invented distribution at all. Pull a real hour out of the 8760 on record.

4.1 Proposition 3.4 — the geometry of diversification

This is the result I most want you to actually *see*, because it looks like algebra and is really a picture.

Each tap has $CV = 1$ (as volatile as it is large). But the *island total* does not. The paper says

$$CV(\text{total}) = \frac{\sqrt{\sum_i \mu_i^2}}{\sum_i \mu_i} = \frac{\|\mu\|_2}{\|\mu\|_1}.$$

Two ways to measure how far you have travelled across a city.

- $\|\mu\|_1$ — **the walk**. You go three blocks east, then four blocks north. Distance travelled: $3 + 4 = 7$ blocks. Add up every leg.
- $\|\mu\|_2$ — **the helicopter**. Straight line from start to finish: $\sqrt{3^2 + 4^2} = 5$ blocks.

The ratio $\|\mu\|_2/\|\mu\|_1 = 5/7$ is a measure of **how many different directions your trip used**.

Now read the two extremes off the picture.

	All load at one bus	Load spread evenly over n
The trip	one straight leg	a long zigzag
Helicopter vs walk	identical	helicopter much shorter
Ratio	1	$1/\sqrt{n}$
Meaning	no cancellation	maximum cancellation

So the bound in the paper,

$$\frac{1}{\sqrt{n}} \leq \text{CV}(\text{total}) \leq 1,$$

is just the statement that **the helicopter never beats the walk, and never loses by more than $\sqrt{n}$** . That is the Cauchy–Schwarz inequality, and that is all it is.

Read the consequence carefully, because it is the opposite of the usual intuition. The volatility of the island total is **not** a property of the distribution you assumed. It is a property of **how concentrated the load is**. O‘ahu gets $\text{CV} = 0.534$ and IEEE-14 gets 0.444 from the *identical* demand model—because two O‘ahu buses carry 72.4 % of the island, so there is less for the randomness to cancel against. Concentrated load diversifies less. Geography, not statistics.

4.2 Proposition 3.6 — the knob that does two things at once

Model B has a knob w : how much of each tap’s behaviour comes from the shared island-wide shock rather than its own private noise.

Rolling dice for each tap’s demand.

- $w = 0$: every tap rolls **its own** die. Independent.
- $w = 1$: one die is rolled **once** and every tap copies it. Perfect lockstep.
- $w = 0.7$: each tap’s number is 70% the shared die, 30% its own.

The paper derives, for any two taps $i \neq j$:

$$\text{CV}(D_i) = \sqrt{w^2 + (1-w)^2}, \quad \text{Corr}(D_i, D_j) = \frac{w^2}{w^2 + (1-w)^2}.$$

At the value used, $w = 0.7$: correlation = 0.845, nodal CV = 0.762.

Two things here caught us out, and both are stated in the paper as corrections.

One: w is **not the correlation**. Setting $w = 0.7$ gives correlation 0.845, not 0.7. It is a *mixing weight*—how much of the recipe comes from the shared jar—and the

correlation it produces is a different number. The earlier write-up called it a correlation. That was wrong, and it is corrected.

Two: turning the knob moves two things at once. Look at the CV formula as w goes $0 \rightarrow 1$:

$$1 \longrightarrow \frac{1}{\sqrt{2}} \approx 0.707 \longrightarrow 1.$$

It **dips in the middle**. Averaging two independent dice is steadier than either one alone—the same reason two coin flips vary less than one. So moving w off zero *raises* correlation and *lowers* each tap’s volatility, simultaneously.

That means Model B is **not a clean experiment about correlation**. It is a joint sensitivity, and the paper labels it as one rather than pretending otherwise.

5 The geometry of failure (paper §4)

Now back to Figure 1, with the results attached.

5.1 Proposition 4.1 — no holes

The claim: walk out along a fixed heading, and the stretch you can serve is one unbroken run from zero out to the fence. You never get a gap—servable, then not, then servable again.

Suppose you have a working setup at full demand: pumps at certain levels, water moving down the pipes, every tap satisfied.

Now **shrink the whole thing by the same factor**—a scale model. Every pump at 60%, every flow at 60%, every tap at 60%.

Check the rig: every pump is running *lower* than before, so still under its plate. Every pipe carries *less*, so still under its rating. And the books still balance, because halving everything on both sides of an equation keeps it true.

That is the proof. Every constraint in the model is a straight-line relationship, and every limit is a ceiling—so scaling a valid answer *down* can never break it. Hence the servable stretch runs from 0 out to some furthest point, with nothing missing in between. The paper calls that furthest point $T^*(\sigma)$: **how far the fence is, on the heading you’re facing**.

5.2 Corollary 4.2 — adequacy is about which way you face

Once you have the fence distance T^* , the entire adequacy question reduces to one comparison:

You must shed load $\iff$ how far you walked $>$ how far the fence is on your heading

$$\|D\|_1 > T^* \left(\frac{D}{\|D\|_1} \right)$$

The bar you must clear is **not a fixed number**. It moves depending on which way you face. That is what “adequacy is angular” means, and the numbers show how much it moves:

Fence distance T^* , over random headings	MW
Worst heading found	409
1st percentile	592
Median heading	1028
99th percentile	1238
Best heading found	1496
The heading real demand actually walks	1082

Read the last two lines together. The real heading is a *comparatively lucky* one: 64.7 % of random headings have their fence closer in than the real one does.

5.3 Proposition 4.5 — and here is where I was wrong

I want you to see this one as a mistake being caught, because that is what it was.

My guess: congestion would also be angular. The reasoning felt airtight—a line gets full when power has to *cross* it, and whether power crosses it depends on *where* the demand sits. So the congestion fence should be lumpy too.

The test that killed it. Sort every scenario by total demand into bands. Then, *within a single band*, compare the two demand models. If congestion were about direction, the models would disagree inside a band, because they face different ways. If it were about magnitude, they would agree, because inside a band they walked the same distance.

Total demand band	Model A congests	Model C congests
600–700 MW	93.4%	100%
700–800 MW	99.7%	100%
800–900 MW	100%	100%

They agree. Conditioning on distance walked wipes out nearly all the difference between two radically different demand models. Congestion is *radial*—a threshold on how far you walked, roughly regardless of heading.

The clinching number: one single threshold, $t_c = 585$ MW, reproduces *both* models' congestion rates.

	Predicted by “walked past 585 MW?”	Actually observed	Error
Model A (exponential)	59.0%	58.4%	0.6 pts
Model C (real hours)	92.0%	92.5%	−0.5 pts

Two models whose congestion rates differ by **34 percentage points**, both predicted to within **half a point** by one number. The 34-point gap was never about direction at all—it is just that the exponential spends 42.8 % of its time loafing around *inside* the round fence, while real demand is almost always outside it.

In one sentence, no symbols: why does sorting scenarios by total demand first, and only then comparing the two models, settle whether a failure mode is about distance or about direction?

5.4 Theorem 4.6 — why the real island never sheds

Go back to the bold arrow in Figure 1.

Real O‘ahu demand, as we model it, **always walks the exact same heading**. The neighbourhood shares never change—Honolulu is always 39.9% of the island, Ewa-Central always 32.5%. Only the *distance* varies, hour to hour.

So the question collapses. There is only one fence distance that matters: 1082 MW, the fence on that one heading. And in 8760 recorded hours, the furthest the island ever walked was 1054.3 MW.

$$1082.0 \text{ MW fence} - 1054.3 \text{ MW peak} = \mathbf{27.7 \text{ MW of margin} = 2.63\%}$$

Zero shedding is a real result. But it is a *2.63% margin*, not a comfortable one—and that is the number worth quoting, not the zero.

Compare the margin you would quote if you only looked at the nameplate on the generators:

$$\frac{1508 \text{ MW} - 1054 \text{ MW}}{1054 \text{ MW}} = 43\% \quad \text{versus} \quad 2.63\%.$$

The gap is 426 MW of generation that exists but sits **stranded behind one 400 MW corridor**—Ewa-West to Ewa-Central, which under real load runs at 97.7% of its rating on average.

A megawatt you cannot move is not capacity.

6 Counting carefully (paper §5)

6.1 Error bars, and why the textbook formula fails here

We ran 10,000 scenarios and got 41.58% shedding. How much of that last digit is real?

It is a coin-flip problem. Flip 10,000 times, count heads, and ask how far the true rate could be from what you saw. At this sample size the answer is about ± 1 percentage point.

Now the case that breaks the standard formula. Model C shed load in **zero** of 10,000 scenarios.

The textbook interval is $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/N}$. Put $\hat{p} = 0$ in and everything collapses: the interval is $[0, 0]$. That says “**this can never happen**” on the strength of 10,000 trials—which is obvious nonsense. A one-in-a-million event would also show up zero times in 10,000 draws.

The **Wilson interval** (equation 11) handles the edge properly and returns $[0, 0.038\%]$. In words: “*we saw none, and the true rate is below about 1 in 2,600—but it is not necessarily zero.*”

That is why the paper reports zero shedding *and* the 2.63 % margin. The zero is what we measured; the margin is what we know.

6.2 Proposition 5.2 — when the solver is picking, not the physics

Two workers, identical in every way, paid exactly the same wage. You need 100 MW of work done.

Any split works: 100/0, 50/50, 73/27. The **total** is pinned by the job. **Who does what** is not pinned by anything.

Now watch what happens if you write down “worker 1 did 73 MW” and call it a finding. You have not measured the world. You have measured **which worker your foreman happened to point at first**.

IEEE-14 has exactly this: two generators at $20 \text{ MW}^{-1} \text{ h}$. Probing confirmed one could move 140 MW with no change in total cost whatsoever. The fix is to pay one worker a token extra— 10^{-6} dollars—so the choice becomes definite and reproducible.

Survives the tie	Does not
Total cost	Which of the two tied units ran
Total shedding	Their individual capacity factors
Whether the expensive tier ran	
Whether a line hit its limit	

All three headline answers sit in the left column—so the tie never threatened them. But the per-generator chart did report solver internals as results, and that is now labelled.

7 What the numbers said

7.1 The finding in one table

Same network. Same solver. Same sample size. Only the demand distribution changed.

	Assumed Exp(1)	Real hours	
Shedding required	19.77%	0.00%	↓ to zero
Costliest unit runs	5.48%	0.00%	↓ to zero
A line at its limit	58.40%	92.51%	↑ 34 points

The two failure modes move in opposite directions. That is the whole paper in three rows. The assumption did not make the grid look uniformly scarier or uniformly safer—it moved the danger from one place to another.

7.2 Why each half happens

Adequacy vanishes	risk	The exponential wanders in random directions, and most directions have their fence closer in than the real heading does. It also throws demand very far out sometimes. Both together mean it crosses the lumpy fence often. Real demand walks one lucky heading and stops just short.
Congestion rises	risk	Congestion is just “did you get past 585 MW.” The exponential spends 42.8 % of its time below that line. Real load almost never drops that low—its floor is 472.6 MW and it sits in the stressed band nearly all the time.

The lesson that generalises beyond this grid: people often defend a fat-tailed demand assumption as “being conservative.”

On this evidence that defence **holds for adequacy and fails for congestion**. You cannot sign the direction of the error just by noting that your assumed distribution has a heavier tail. It depends on whether the failure you care about is triggered by *crossing a line* or by *sitting inside a band*.

7.3 What we are still not sure about

The honest limitations, in the order they would bite:

- L1. Model C fixes the heading by construction.** Real demand surely wobbles a little in direction; we gave it none. Since the real heading is a lucky one, allowing wobble would probably produce a small but nonzero shedding probability. *The zero is fragile; the 2.63 % margin is the honest headline.*
- L2. Bus loads are guessed from population.** Honolulu’s hotels, offices and military bases do not draw power in proportion to who sleeps there. The true heading may be less lucky than the one we used.
- L3. The network is stylised.** Nine buses on a published ring layout, with assumed ratings. The *ranking* of corridors is solid; the exact percentages inherit the assumptions.
- L4. No unit commitment, no contingencies.** Generators can switch off freely, and nothing ever breaks. Given that the finding is one binding corridor, asking “what if that corridor trips?” is the most valuable next step.

8 Symbol dictionary

Every symbol in the formal paper, with the object it stands for.

Symbol	Object	Job
D_i	how far tap i is open	demand at bus i (MW)
μ_i	the tap's <i>usual</i> position	average demand at bus i
S_i	the overflow drain at tap i	demand not served
P_g	pump g 's throttle	generator output
$p_g^{\max}$	the pump's rating plate	generator capacity
c_g	the pump's price tag	marginal cost, $\text{MW}^{-1} \text{h}$
θ_i	pressure at junction i	voltage phase angle
F_ℓ	flow down pipe ℓ	branch power flow (MW)
$\bar{F}_\ell$	pipe ℓ 's rating	thermal limit
VOLL	the fine per litre spilled	value of lost load
λ_i°	value of one more litre at tap i	locational marginal price / dual
$T = \ D\ _1$	how far you walked	total island demand
$\sigma = D/T$	which way you face	the mix across neighbourhoods
$T^*(\sigma)$	how far the fence is, facing σ	deliverable capacity on that heading
$\bar{\sigma}$	the heading real demand walks	population-share direction
t_c	the round fence	congestion threshold, 585 MW
ρ	how hard the grid is working	mean demand $\div$ nameplate
CV	how wide the sand pile spreads	std. deviation $\div$ mean
n	how many taps carry load	number of load buses
N	how many times you rolled	Monte Carlo replications (10,000)
w	how much of the recipe is shared	mixing weight in Model B
$\hat{p}$	the tally you counted	estimated probability

9 The five teach-backs, collected

If you can answer these out loud without notes, you own the paper.

1. Using the full-pipe picture and no symbols: why is “set the penalty higher than the most expensive generator” *not* a safe rule?
2. Using the walk-versus-helicopter picture: why does the same demand model produce a more volatile island total on O‘ahu than on IEEE-14?
3. Why does sorting scenarios by total demand *first*, and only then comparing the two models, settle whether a failure mode is about distance or about direction?
4. Using the two-fences field: why can real O‘ahu demand hit the round fence almost every hour and never once reach the lumpy one?
5. Why is 2.63 %, rather than 43 %, the reserve margin worth quoting—and where did the other 426 MW go?

*Companion to set8.pdf. Same results, same numbers, same order.
If an anchor here does not click, say which one—it gets rebuilt from a different object.*